

GARMENT WARPING & FITTING TECHNIQUES FOR VIRTUAL TRY-ON SYSTEMS

Research Report

Member 3 – Garment Warping & Fitting Research

AI/ML Internship – AR/VR Virtual Try-On System

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1. INTRODUCTION

Virtual Try-On (VTON) technology enables users to visualize garments on their own body without physically wearing them. This technology has become increasingly important in e-commerce and fashion retail because it improves customer experience and reduces product returns.

The primary challenge in virtual try-on systems is adapting clothing to different body shapes, poses, and sizes while maintaining realistic appearance and texture. To solve this problem, modern systems use garment warping and fitting techniques such as Thin Plate Spline (TPS), Flow-Based Warping, CP-VTON, VITON-HD, and Geometric Matching Modules (GMM).

These methods help align and transform garments according to the target person's body structure while preserving the garment's visual characteristics.

2. THIN PLATE SPLINE (TPS)

Overview

Thin Plate Spline (TPS) is a classical geometric transformation technique widely used in computer vision.

TPS transforms clothing images based on corresponding control points between the garment and the target body.

Working Principle

1. Detect body landmarks.
2. Select garment control points.
3. Compute TPS transformation parameters.
4. Warp garment image.
5. Generate aligned garment output.

Advantages

- * Smooth transformation
- * Fast execution
- * Low computational cost
- * Easy implementation

Limitations

- * Less effective for complex clothing folds
- * Limited realism for extreme poses

Architecture Diagram

Garment Image



Control Point Selection



TPS Transformation



Warped Garment



Virtual Try-On Output



3. FLOW-BASED WARPING

Overview

Flow-based warping predicts the movement of every garment pixel using deep learning.

Unlike TPS, which uses geometric control points, flow-based approaches generate dense motion fields.

Working Principle

1. Input garment image.
2. Input person image.
3. Extract features.
4. Predict flow field.
5. Move garment pixels accordingly.
6. Generate final fitted garment.

Advantages

- * Handles complex body poses
- * Better cloth deformation
- * More realistic fitting

Limitations

- * Requires powerful hardware
- * Computationally expensive

Architecture Diagram

Person Image + Garment Image

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CNN Encoder

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Flow Prediction

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4. CP-VTON

Full Form

Characteristic-Preserving Virtual Try-On Network

Overview

CP-VTON was one of the first successful deep learning virtual try-on systems.

It consists of:

- * Geometric Matching Module (GMM)
- * Try-On Module

Advantages

- * Preserves garment texture
- * Good garment alignment
- * Strong research benchmark

Limitations

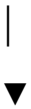
- * Lower output resolution
- * Reduced detail preservation

Architecture Diagram

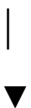
Person Representation



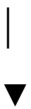
Garment Image



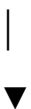
Geometric Matching Module



Warped Garment



Try-On Module



Final Output

5. VITON-HD

Full Form

High-Definition Virtual Try-On Network

Overview

VITON-HD improves upon CP-VTON by producing high-resolution virtual try-on images.

It generates realistic textures, folds, and garment boundaries.

Stages

Stage 1

Garment Alignment

Stage 2

High Resolution Synthesis

Stage 3

Refinement Network

Advantages

- * Very high image quality
- * Better texture preservation
- * Realistic garment appearance

Limitations

- * Requires more GPU memory
- * Longer training time

Architecture Diagram

Person Image



Garment Image



Garment Alignment



High Resolution Synthesis



Refinement Network



Final HD Output

6. GEOMETRIC MATCHING MODULE (GMM)

Overview

GMM learns the geometric transformation required to align clothing with the target body.

The module predicts TPS parameters automatically.

Advantages

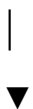
- * Accurate alignment
- * Automatic learning
- * Improved fitting quality

Architecture Diagram

Person Features

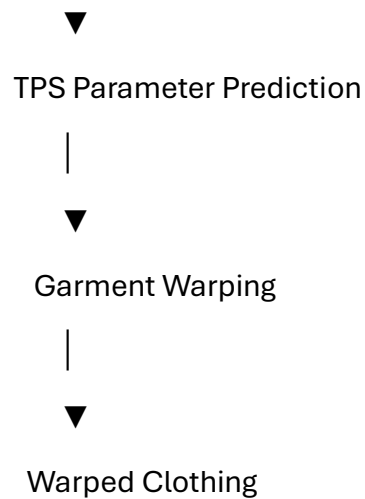


Garment Features



Feature Correlation





7. COMPARISON TABLE

Technique	Accuracy		Quality		Speed	Complexity	Best Use Case
TPS	Medium		Medium		Fast	Low	Basic Warping
Flow-Based Warping	High		High		Medium	High	Complex Poses
GMM	High	High	Medium		Medium		Garment Alignment
CP-VTON	High	High	Medium		Medium		General Virtual Try-On
VITON-HD	Very High		Very High		Slower	High	Commercial Applications

8. COMPLETE VIRTUAL TRY-ON WORKFLOW

Step 1 – User Uploads Image

User Image





Image Acquisition

Step 2 – Pose Detection

User Image



Pose Estimation

(MediaPipe/OpenPose)



Body Landmarks

Step 3 – Human Segmentation

Body Image



Segmentation Network



Body Regions

Step 4 – Garment Processing

Garment Image





Feature Extraction

Step 5 – Garment Warping

Body Features



Garment Features



TPS / GMM / Flow-Based Warping



Warped Garment

Step 6 – Image Synthesis

Warped Garment



Image Generation Network



Virtual Try-On Image

Step 7 – Final Output

Virtual Try-On Result



Display To User

Overall Workflow Diagram

User Image



Pose Detection



Human Segmentation



Garment Processing



Garment Warping

(TPS / GMM / Flow-Based)



Image Synthesis



Final Virtual Try-On Output

9. APPLICATIONS

- * Fashion E-Commerce
- * Online Shopping Platforms
- * AR Shopping Applications
- * Fashion Design
- * Metaverse Avatars
- * Gaming Industry
- * Retail Stores

10. FUTURE SCOPE

Future virtual try-on systems are expected to incorporate:

- * Real-time 3D garment simulation
- * Physics-based cloth modeling
- * AI-generated avatars
- * Diffusion models
- * AR smart glasses integration
- * Personalized fashion recommendations

11. CONCLUSION

Garment warping and fitting techniques form the foundation of modern virtual try-on systems. TPS provides efficient geometric transformations, while Flow-Based Warping, CP-VTON, and VITON-HD produce increasingly realistic garment fitting results. Among these techniques, VITON-HD currently offers the best visual quality, whereas CP-VTON provides a balanced solution between accuracy and computational efficiency.
